

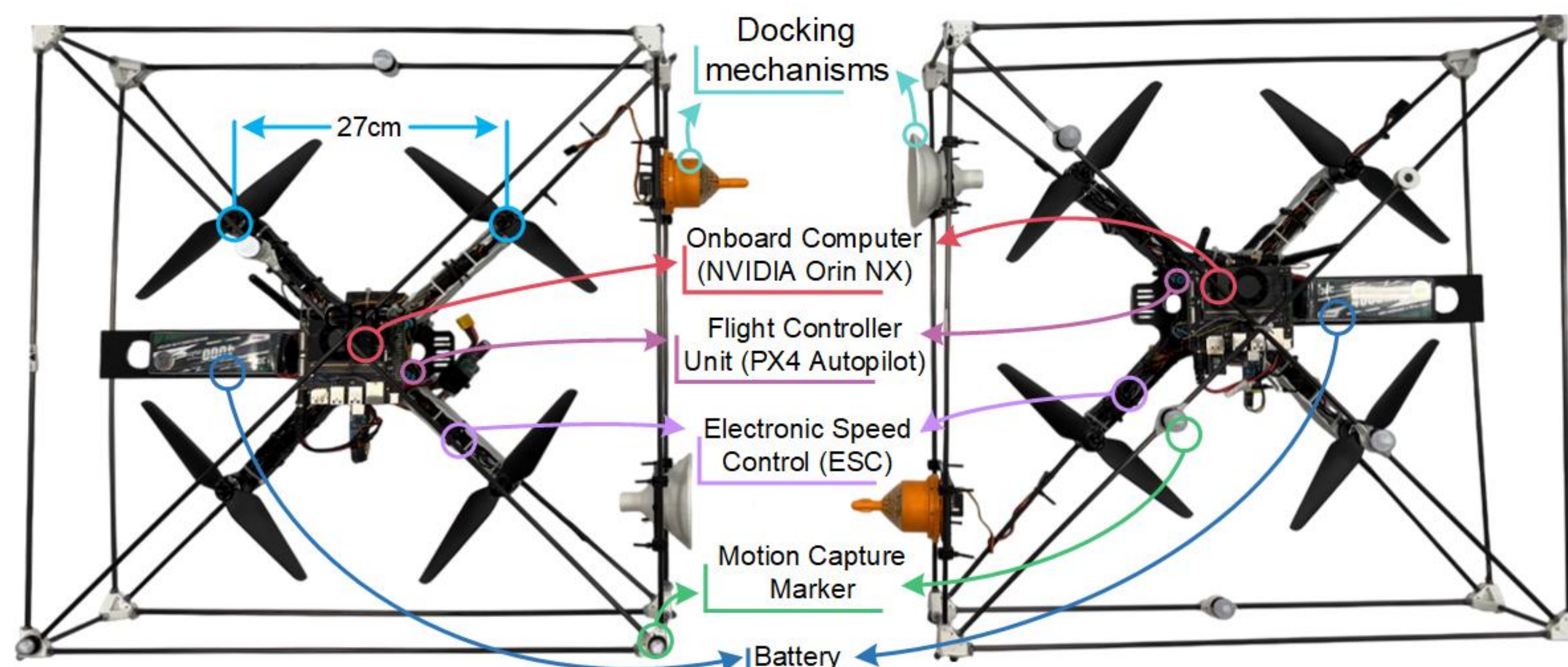
# MARS-FTCP: Robust Fault-Tolerant Control and Agile Trajectory Planning for Modular Aerial Robot Systems

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## Motivations

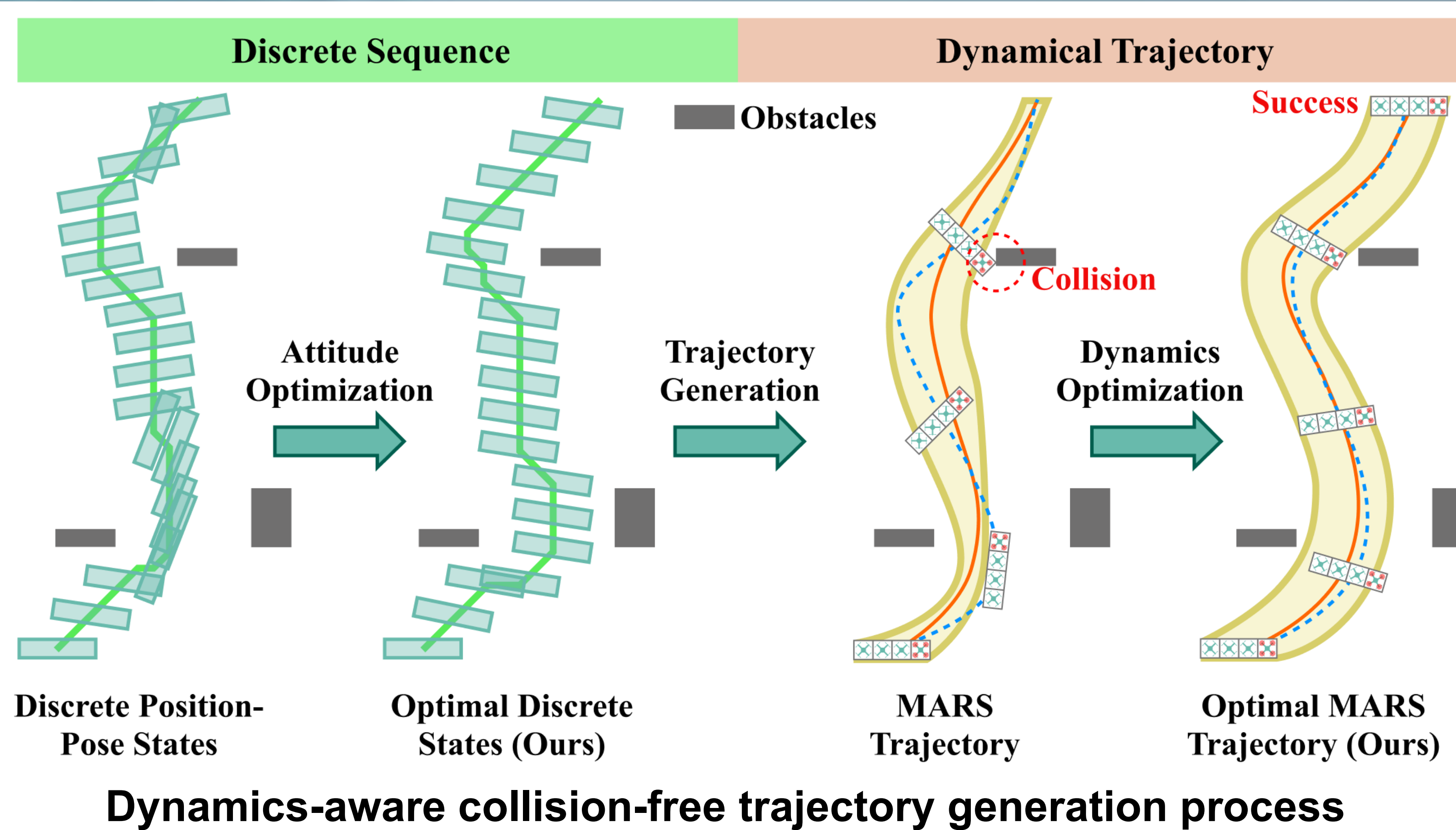
- ❑ **MARS capability:** adaptive reconfiguration for diverse tasks and resilient operation under faults.
- ❑ **Fault tolerance:** maintain stability and functionality under unit and rotor failures.
- ❑ **Dynamic feasibility:** neglect of dynamic asymmetry limits real-world trackability.
- ❑ **Objectives:** collision-free flight and reduced tracking errors.



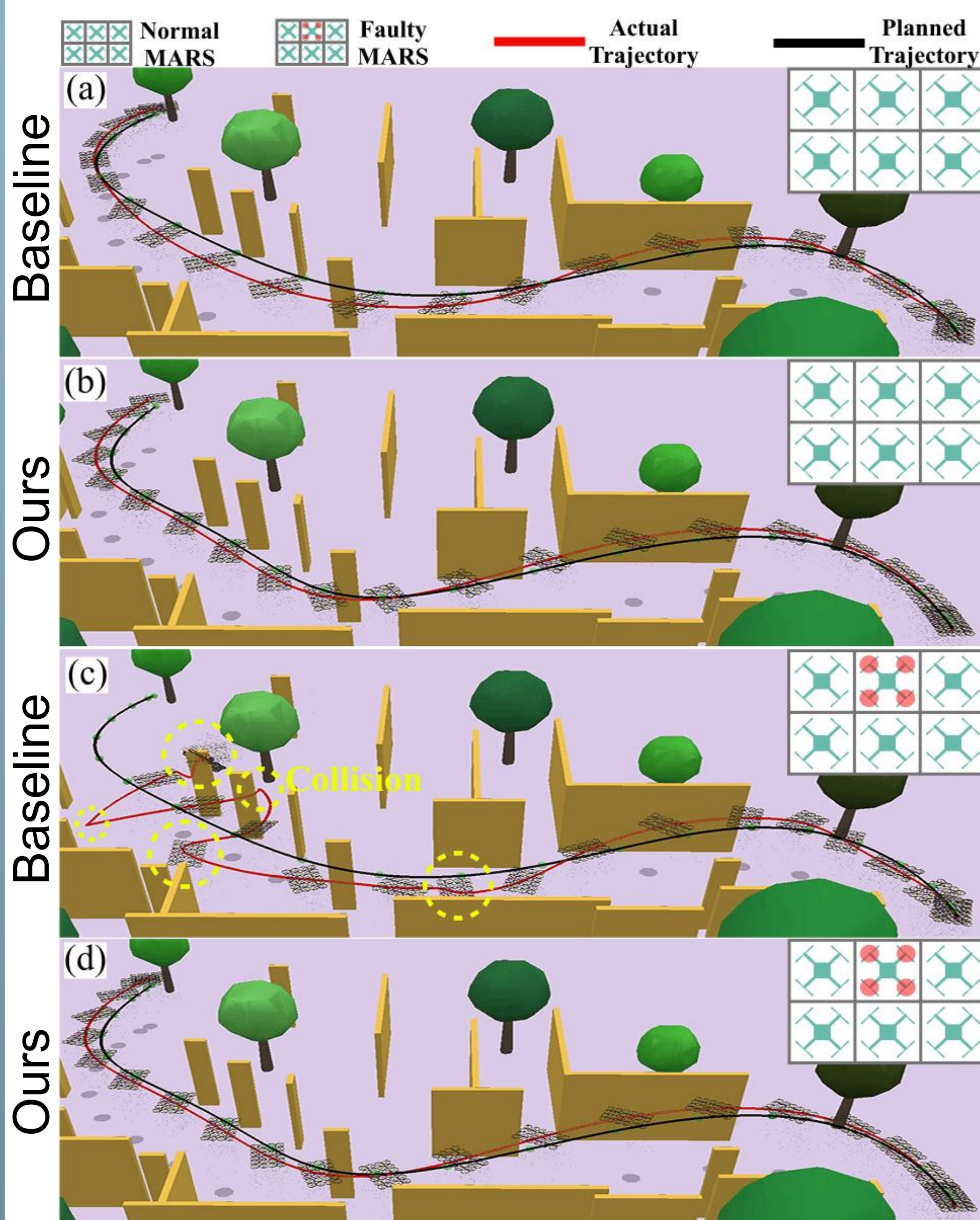
Modular Aerial Robot Systems (MARS)

## Methodology

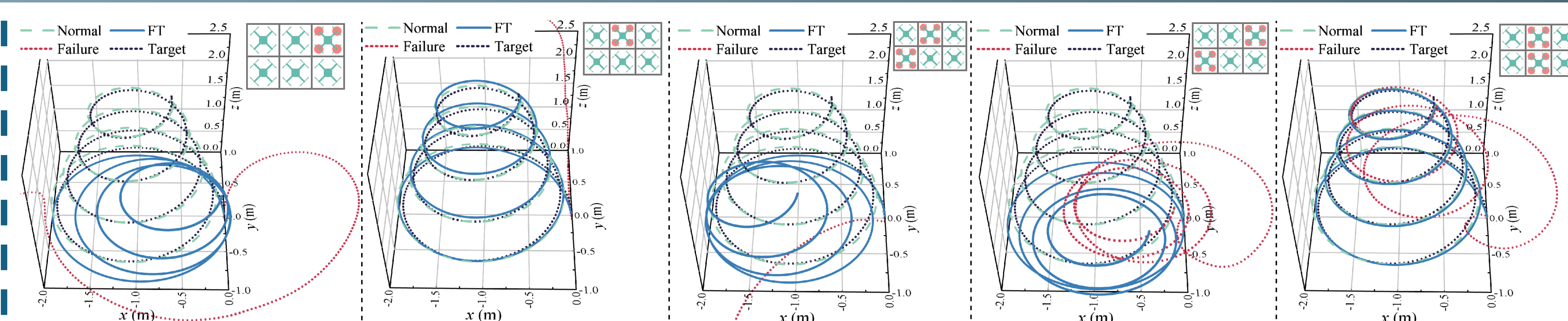
- ❑ **Fault tolerance control:**
  - (1) **Units failure:** reallocate force and torque among all normal units.
  - (2) **Rotor failure:** 1) Partial reallocation: MARS global reallocation; 2) Full reallocation: Units local fault tolerance + MARS global reallocation.
- ❑ **Collision-free trajectory planning:**
  - (1) **Discrete sequence:** dynamically feasible A\* generates optimized paths.
  - (2) **Dynamic trajectory:** generates smooth, dynamically feasible, and collision-free flight paths.



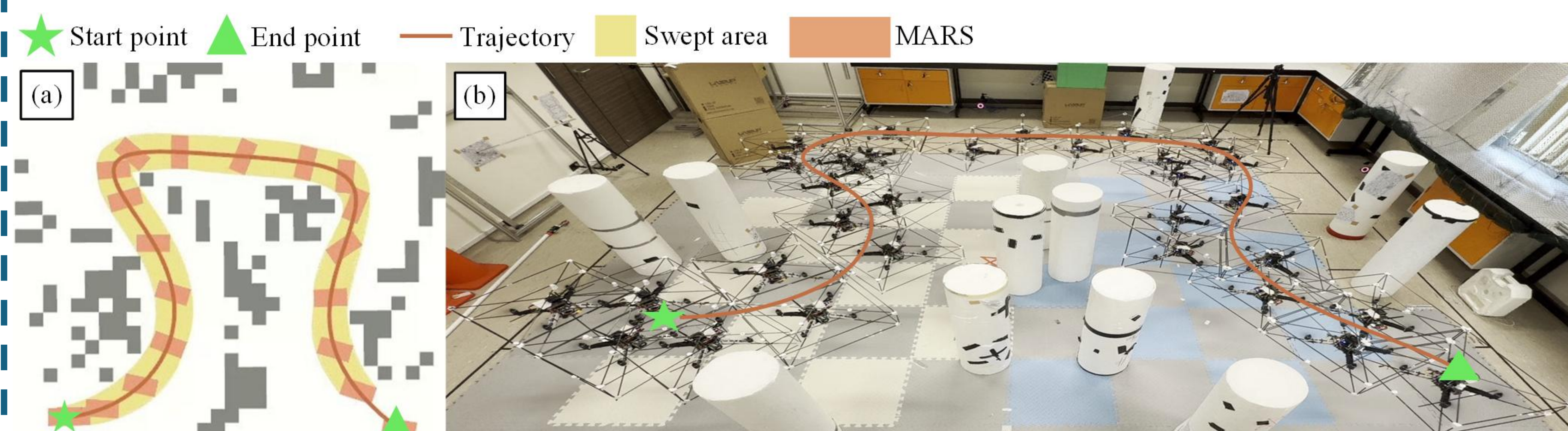
## Experiment Result



① Simulation



② Fault tolerance control in different configurations



③ Real-world experiment

Trajectory error analysis: collision-free path planning

| Shape (Fault ID) | Baseline [6] (m) | Ours (m) | Improvement |
|------------------|------------------|----------|-------------|
| 4 × 1 (Normal)   | 0.5536           | 0.2849   | 48.5%       |
| 4 × 1 (No. 3)    | 0.7499           | 0.3039   | 59.5%       |
| 4 × 1 (No. 4)    | 0.9923           | 0.3068   | 69.1%       |
| 3 × 2 (Normal)   | 1.1777           | 0.3216   | 72.7%       |
| 3 × 2 (No. 2)    | 1.7007           | 0.3486   | 79.5%       |
| Average          | 1.0348           | 0.3132   | 69.7%       |

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